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Factuality Verification and Self-Reflective Steering in Cognitive AI Agents

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ABSTRACT

Generative cognitive agents frequently suffer from subtle hallucinations when synthesizing complex multi-source literature. We develop a self-reflective steering protocol that evaluates intermediate chain-of-thought tokens against verified factual triples before response generation. Our method incorporates an adversarial critic agent that formulates counter-factual probes, penalizing unverified claims during beam search. Extensive benchmarks demonstrate a 62% reduction in hallucinated assertions and an 18-point increase in factual attribution score. A notable drawback is the 2.3x increase in inference computation and latency, limiting its viability in latency-critical interactive dialogues.

Keywords— Artificial Intelligence, Cognitive Agents, Factuality, Self-Reflection

INTRODUCTION

As autonomous AI assistants assume central roles in scholarly research and medical diagnosis, factual fidelity is non-negotiable. Standard autoregressive decoding lacks an internal feedback loop for factual validation, producing fluent yet erroneous claims.

METHODOLOGY

We formulate a dual-agent architecture comprising a generator and an internal adversarial critic. The critic verifies claims against a structured facts repository, computing a semantic divergence loss that steers subsequent decoding steps.

RESULTS

In scientific QA benchmarks, self-reflective steering boosted factual correctness from 74.2% to 92.5%, significantly exceeding baseline prompt-engineering and self-consistency mechanisms.

LIMITATIONS

The verification loop incurs substantial computational overhead due to recursive self-critique calls, resulting in a twofold increase in token generation time.

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